

Exact Positive-Diagonal Matignon Stability in Dimension Three

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Abstract

We study positive-diagonal multiplicative stability for commensurate fractional-order systems in the Matignon region. For real 3×3 matrices on the full-dimensional strict- $P(-A)$ stratum, four orbit invariants reduce the complete positive-diagonal orbit to a two-dimensional simplex and fractional D-stability is equivalent to $\kappa < T_\alpha(\beta)$. The threshold reduces exactly to Cain's classical boundary at integer order; for $2/3 < \alpha < 1$, the strict band $T_1(\beta) < \kappa < T_\alpha(\beta)$ is genuinely fractional and full dimensional. The threshold problem is globally strictly convex in logit coordinates, has a unique smooth optimizer, and admits a sharp classical-order expansion. Dimension three is the smallest real dimension in which the fractional difference from Hurwitz D-stability has nonempty full-dimensional interior. For three-species generalized Lotka–Volterra systems, the invariants become abundance-independent reciprocal-pair and directed-cycle feedback coordinates. High-precision and interval-certified computations independently corroborate the analytic results.

Keywords: D-stability; fractional-order systems; Matignon stability; P-matrices; positive diagonal scaling; generalized Lotka–Volterra systems

MSC: 15A18; 34A08; 93D20

1. Introduction

Classical D-stability asks whether a real matrix remains stable after multiplication by every positive diagonal matrix. In dimension three the Hurwitz problem has an exact solution due to Cain [1], while generalized matrix-stability frameworks extend the spectral target from a half-plane to more general regions [2,3]. For a commensurate Caputo system of order $0 < \alpha < 1$, asymptotic stability is instead governed by the Matignon sector

$$\Sigma_\alpha = \{z \neq 0 : |\arg z| > \alpha\pi/2\}$$

[4,5]. Fixed-polynomial fractional root criteria and cyclic-network secant conditions are known [6–8]. In directly preceding ecological work, Alraddadi and Alharthi obtained exact order-dependent stability atlases for a Caputo Allee predator–prey model and identified purely fractional stabilization, in which the fractional equilibrium is stable despite a right-half-plane Jacobian spectrum [9]. That model-specific result motivates the structural question here, but it does not eliminate the universal positive-diagonal multiplier for a generic real 3×3 matrix.

We solve that problem on the full-dimensional strict- $P(-A)$ stratum. Four positive-left-diagonal orbit invariants,

$$(\beta_{12}, \beta_{13}, \beta_{23}, \kappa),$$

Received:

Accepted:

Published:

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reduce the complete positive-diagonal orbit to the open simplex, and the all-diagonal Matignon condition becomes the scalar inequality

$$\kappa < T_\alpha(\beta).$$

At $\alpha = 1$, T_α reduces exactly to Cain's threshold. For $2/3 < \alpha < 1$, the strict interval

$$T_1(\beta) < \kappa < T_\alpha(\beta)$$

is therefore an exact genuinely fractional D-stable band.

The paper makes four main contributions. First, dimension two is classified exactly and cannot support a full-dimensional genuinely fractional difference class. Second, the real- 3×3 positive-diagonal Matignon problem is reduced to the exact threshold T_α , with a complete characterization of the interior. Third, the threshold objective is globally strictly convex in logit coordinates, giving a unique smooth optimizer, strict monotonicity in α , and a sharp classical-order expansion. These results imply that dimension three is the smallest real dimension with a robust genuinely fractional separation from Hurwitz D-stability. Fourth, for three-species generalized Lotka–Volterra systems the same invariants become abundance-independent reciprocal-pair and directed-cycle feedback coordinates.

The contribution is not a new definition of fractional D-stability. It is an explicit low-dimensional solution within the generalized-D-stability framework. Kushel and Pavani [3] give an abstract forbidden-boundary framework for positive diagonal multipliers. Here the multiplier is eliminated exactly for the generic real three-dimensional Matignon problem. On the symmetric slice, our formula recovers Siami's single-cycle result [8]. Relative to the preceding ecological stability-atlas study [9], the present paper moves from a fixed nonlinear model and its parameter atlas to an exact matrix-orbit theorem quantified over every positive diagonal multiplier.

Throughout, *genuinely fractional* means stable in the Matignon sense for every positive diagonal multiplier but not Hurwitz D-stable. This orbit-level definition is stronger than exhibiting one matrix that is fractionally stable and integer-order unstable only at $D = I$.

Section 2 gives the orbit reduction, Section 3 treats dimension two, Section 4 proves the exact three-dimensional criterion, and Section 5 develops the convex threshold geometry and minimum-dimension consequence. Section 6 gives the GLV interpretation, Section 7 summarizes independent computational certification, and Section 8 positions the result relative to prior work and states its limitations.

2. Positive-diagonal Matignon stability and orbit reduction

For the commensurate Caputo system

$${}^C D_t^\alpha x = Ax, \quad 0 < \alpha < 1,$$

asymptotic stability is equivalent to

$$\sigma(A) \subset \Sigma_\alpha := \{z \neq 0 : |\arg z| > \alpha\pi/2\}$$

[4,5]. Define

$$\mathcal{F}_\alpha^{(n)} = \{A : \sigma(DA) \subset \Sigma_\alpha \forall D \succ 0 \text{ diagonal}\},$$

$$\mathcal{D}_H^{(n)} = \{A : DA \text{ Hurwitz} \forall D \succ 0\}, \quad \mathcal{P}_\alpha^{(n)} = \mathcal{F}_\alpha^{(n)} \setminus \mathcal{D}_H^{(n)}.$$

All interiors are taken in the Euclidean topology of $\mathbb{R}^{n \times n}$. Because

$$DA = D(AD)D^{-1},$$

left and right positive-diagonal multiplication are spectrally equivalent. Our Hurwitz sign convention corresponds to Cain's positive-stability convention after replacing his matrix M by $-A$ [1].

A real matrix is a P-matrix if every principal minor is positive and a P_0 -matrix if every principal minor is nonnegative. Assume $-A$ is strict P and, in dimension three, write

$$p_i = -a_{ii} > 0, \quad m_{ij} = \det A[\{i, j\}] > 0, \quad q = -\det A > 0.$$

Define the positive-left-diagonal invariants

$$\beta_{12} = \frac{m_{12}}{p_1 p_2}, \quad \beta_{13} = \frac{m_{13}}{p_1 p_3}, \quad \beta_{23} = \frac{m_{23}}{p_2 p_3}, \quad \kappa = \frac{q}{p_1 p_2 p_3}.$$

Lemma 1 (Positive-diagonal orbit reduction). *Let*

$$D = \text{diag}(d_1, d_2, d_3) \succ 0, \quad s = \sum_{i=1}^3 p_i d_i, \quad x_i = \frac{p_i d_i}{s}.$$

Then $x \in \Delta_2^\circ = \{x_i > 0 : \sum_i x_i = 1\}$, and

$$\det(\lambda I - DA) = \lambda^3 + s\lambda^2 + s^2 B_\beta(x)\lambda + s^3 \kappa x_1 x_2 x_3,$$

where

$$B_\beta(x) = \beta_{12} x_1 x_2 + \beta_{13} x_1 x_3 + \beta_{23} x_2 x_3.$$

Conversely every $x \in \Delta_2^\circ$ arises from a positive diagonal D , uniquely up to common positive scale.

Proof. The characteristic coefficients are

$$s = \sum_i p_i d_i, \quad b_D = \sum_{i < j} m_{ij} d_i d_j, \quad c_D = q d_1 d_2 d_3.$$

Substituting $d_i = s x_i / p_i$ gives

$$b_D = s^2 B_\beta(x), \quad c_D = s^3 \kappa x_1 x_2 x_3.$$

Conversely $d_i = s x_i / p_i$ reconstructs D for any $s > 0$. $\square$

The radial substitution $\lambda = sz$ shows that s changes only eigenvalue magnitudes, not arguments. Thus the full positive-diagonal orbit modulo common scale is exactly the open simplex. The four ratios (β, κ) are unchanged by positive left-diagonal scaling.

3. Dimension two: exact obstruction

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Theorem 1 (Exact 2×2 classification). *For every $0 < \alpha < 1$,*

$$A \in \mathcal{F}_\alpha^{(2)} \iff \det A > 0, \quad a \leq 0, \quad d \leq 0.$$

Proof. For $D = \text{diag}(x, y) \succ 0$,

$$\text{tr}(DA) = xa + yd, \quad \det(DA) = xy \det A.$$

If $\det A \leq 0$, either zero is an eigenvalue or DA has a positive real eigenvalue. If $a > 0$, fixing $x > 0$ and sending $y \downarrow 0$ makes DA approach a matrix with simple positive eigenvalue xa ; continuity gives a positive eigenvalue for small $y > 0$. Hence $a \leq 0$, and similarly $d \leq 0$.

Conversely, if $\det A > 0$ and $a, d \leq 0$, every DA has positive determinant and non-positive trace. Its real eigenvalues are therefore negative; a nonreal conjugate pair has nonpositive real part and hence arguments of magnitude at least $\pi/2 > \alpha\pi/2$. $\square$

Classical Hurwitz D-stability imposes the strict trace inequality

$$xa + yd < 0 \quad \forall x, y > 0,$$

so under the theorem's conditions it fails exactly when $a = d = 0$. Therefore

$$\mathcal{P}_\alpha^{(2)} = \left\{ \begin{pmatrix} 0 & b \\ c & 0 \end{pmatrix} : bc < 0 \right\}.$$

Corollary 1 (No robust two-dimensional separation). *For every $0 < \alpha < 1$,*

$$\text{int}_{\mathbb{R}^4} \mathcal{P}_\alpha^{(2)} = \emptyset.$$

Proof. The set is contained in the codimension-two subspace $a = d = 0$. $\square$

4. Exact three-dimensional Matignon threshold

Let $-A$ be strict P and use the notation of Lemma 1. For $D \succ 0$,

$$\det(\lambda I - DA) = \lambda^3 + a_D \lambda^2 + b_D \lambda + c_D,$$

with

$$a_D = s, \quad b_D = s^2 B_\beta(x), \quad c_D = s^3 \kappa x_1 x_2 x_3.$$

4.1. Exact cubic boundary

Lemma 2 (Exact cubic Matignon boundary). *Let*

$$p(\lambda) = \lambda^3 + a\lambda^2 + b\lambda + c, \quad a, b, c > 0,$$

and $\pi/3 < \theta < \pi/2$. Set

$$u = \cos \theta, \quad K = 1 - 4u^2 > 0,$$

$$r_\theta(a, b) = \frac{ua + \sqrt{u^2 a^2 + Kb}}{K},$$

and

$$H_\theta(a, b) = r_\theta(a, b)^2 (a + 2u r_\theta(a, b)).$$

Then all roots satisfy

$$|\arg \lambda| > \theta$$

if and only if

$$c < H_\theta(a, b).$$

Moreover,

$$H_\theta(sa, s^2 b) = s^3 H_\theta(a, b).$$

Proof. A positive-coefficient cubic has no nonnegative real root. A root can therefore leave the open angular region only through one of the rays $\lambda = re^{\pm i\theta}$, $r > 0$. Substituting $\lambda = re^{i\theta}$ and dividing the imaginary part by $r \sin \theta$ gives

$$Kr^2 - 2uar - b = 0,$$

whose unique positive solution is $r_\theta(a, b)$. The real part then gives

$$c = r^2(a + 2ur) = H_\theta(a, b).$$

Thus there is exactly one positive c at which a ray is touched. For $c > 0$ sufficiently small, one root is a small negative real root and the other two perturb the Hurwitz roots of $\lambda^2 + a\lambda + b$; for $c \rightarrow \infty$, two roots approach the cube-root directions $\pm\pi/3$, which violate $\theta > \pi/3$. Hence stability is exactly $c < H_\theta(a, b)$. Homogeneity follows by $r_\theta(sa, s^2b) = sr_\theta(a, b)$. $\square$

For $2/3 < \alpha < 1$, put

$$\theta = \frac{\alpha\pi}{2}, \quad u = \cos \theta, \quad K = 1 - 4u^2,$$

and define

$$r_\alpha(b) = \frac{u + \sqrt{u^2 + Kb}}{K}, \quad h_\alpha(b) = r_\alpha(b)^2(1 + 2u r_\alpha(b)).$$

4.2. Exact elimination of the diagonal multiplier

Define

$$T_\alpha(\beta) = \min_{x \in \Delta_2^\circ} \frac{h_\alpha(B_\beta(x))}{x_1 x_2 x_3}.$$

The minimum is attained in the interior because the numerator has a positive continuous extension to the closed simplex while $x_1 x_2 x_3 \rightarrow 0$ on its boundary.

Theorem 2 (Exact real-3 $\times$ 3 criterion). *Let $-A$ be strict P and $2/3 < \alpha < 1$. Then*

$$A \in \mathcal{F}_\alpha^{(3)} \iff \kappa < T_\alpha(\beta).$$

Proof. For the simplex coordinate x associated with D ,

$$\det(\lambda I - DA) = \lambda^3 + s\lambda^2 + s^2 B_\beta(x)\lambda + s^3 \kappa x_1 x_2 x_3.$$

Lemma 2 and its homogeneity give

$$\sigma(DA) \subset \Sigma_\alpha \iff \kappa < \frac{h_\alpha(B_\beta(x))}{x_1 x_2 x_3}.$$

Lemma 1 identifies positive diagonal ratios with all $x \in \Delta_2^\circ$, so the inequality holds for every $D \succ 0$ exactly when $\kappa < T_\alpha(\beta)$. $\square$

This is an explicit solution of the real three-dimensional Matignon instance of generalized positive-diagonal region stability. The abstract all- D framework itself is known [2,3].

4.3. Interior classification

Lemma 3 (P_0 necessity). If $A \in \mathcal{F}_\alpha^{(3)}$, then

$$-a_{ii} \geq 0, \quad \det A[\{i, j\}] \geq 0, \quad -\det A > 0.$$

Proof. If $a_{ii} > 0$, keep the i -th row scaling fixed and send the other two to zero; the limiting matrix has a positive eigenvalue $d_i a_{ii}$, which persists for nearby positive scalings. If an order-two principal minor is negative, send the remaining row scaling to zero; the limiting 2×2 principal block has eigenvalues of opposite signs. Both contradict Matignon stability. Finally, at $D = I$, the real eigenvalue of a real cubic must be negative, while the product of the remaining two eigenvalues is positive; hence $\det A < 0$. $\square$

Corollary 2 (Strict-P interior). If

$$A \in \text{int } \mathcal{F}_\alpha^{(3)},$$

then $-A$ is strict P.

Proof. Lemma 3 gives nonnegative signed minors. A vanishing order-one minor can be perturbed immediately to the forbidden sign. A singular 2×2 principal block can likewise be perturbed arbitrarily little to negative determinant: use a first-order adjugate perturbation when the block has rank one, and $\text{diag}(t, -t)$ when it is zero. Thus no such equality can occur at an interior point; the determinant inequality is already strict. $\square$

Kellogg's theorem [10] states that every eigenvalue μ of a 3×3 strict P-matrix satisfies

$$|\arg \mu| < 2\pi/3.$$

Applied to $-DA$, it yields

$$|\arg \lambda(DA)| > \pi/3.$$

Theorem 3 (Interior of $\mathcal{F}_\alpha^{(3)}$). For $0 < \alpha \leq 2/3$,

$$\text{int } \mathcal{F}_\alpha^{(3)} = \{A : -A \text{ strict } P\}.$$

For $2/3 < \alpha < 1$,

$$\text{int } \mathcal{F}_\alpha^{(3)} = \{A : -A \text{ strict } P, \kappa < T_\alpha(\beta)\}.$$

Proof. For $\alpha \leq 2/3$, Kellogg gives sufficiency, Corollary 2 gives necessity, and strict P is open.

For $\alpha > 2/3$, necessity follows from Corollary 2 and Theorem 2. Conversely, near any strict-P matrix the invariants vary continuously and remain in a compact positive box. On a sufficiently small parameter neighborhood, $B_\beta(x)$ ranges in a common compact interval $[0, M]$. Since h_α is continuous and strictly positive on this interval,

$$m := \min_{0 \leq b \leq M} h_\alpha(b) > 0, \quad F_{\alpha, \beta}(x) \geq \frac{m}{x_1 x_2 x_3}.$$

Thus all nearby minimizers lie in one compact subset of Δ_2° , and $T_\alpha(\beta)$ is locally continuous. The strict gap $T_\alpha(\beta) - \kappa > 0$ therefore persists under sufficiently small perturbations. $\square$

4.4. Cain endpoint and the exact fractional band

As $\alpha \rightarrow 1^-$,

$$h_\alpha(b) \rightarrow b,$$

so

$$T_\alpha(\beta) \rightarrow T_1(\beta) = \min_{x \in \Delta_2^s} \frac{B_\beta(x)}{x_1 x_2 x_3}.$$

Since

$$\frac{B_\beta(x)}{x_1 x_2 x_3} = \frac{\beta_{23}}{x_1} + \frac{\beta_{13}}{x_2} + \frac{\beta_{12}}{x_3},$$

Cauchy–Schwarz gives

$$T_1(\beta) = \left(\sqrt{\beta_{12}} + \sqrt{\beta_{13}} + \sqrt{\beta_{23}} \right)^2.$$

This is exactly Cain’s real-3 $\times$ 3 D-stability boundary [1].

Lemma 4 (Strict fractional expansion of Cain’s boundary). *For $2/3 < \alpha < 1$,*

$$T_\alpha(\beta) > T_1(\beta) \quad \forall \beta > 0.$$

Proof. For $r = r_\alpha(b)$,

$$b = Kr^2 - 2ur,$$

hence

$$h_\alpha(b) - b = 4u^2r^2 + 2ur(r^2 + 1) > 0.$$

Apply this pointwise to the fractional objective and then minimize. $\square$

Corollary 3 (Exact genuinely fractional band). *For $2/3 < \alpha < 1$,*

$$\text{int } \mathcal{P}_\alpha^{(3)} = \{A : -A \text{ strict } P, T_1(\beta) < \kappa < T_\alpha(\beta)\}.$$

For $0 < \alpha \leq 2/3$,

$$\text{int } \mathcal{P}_\alpha^{(3)} = \{A : -A \text{ strict } P, \kappa > T_1(\beta)\}.$$

5. Convex geometry and the minimum robust dimension

Theorem 2 reduces the all-diagonal problem to

$$T_\alpha(\beta) = \min_{x \in \Delta_2^s} F_{\alpha,\beta}(x), \quad F_{\alpha,\beta}(x) = \frac{h_\alpha(B_\beta(x))}{x_1 x_2 x_3}.$$

We now show that this variational problem is globally well behaved.

5.1. Strict convexity and uniqueness

Parameterize the simplex by

$$x_1 = \frac{e^{y_1}}{e^{y_1} + e^{y_2} + 1}, \quad x_2 = \frac{e^{y_2}}{e^{y_1} + e^{y_2} + 1}, \quad x_3 = \frac{1}{e^{y_1} + e^{y_2} + 1}.$$

Set

$$L(y) = \log(e^{y_1} + e^{y_2} + 1),$$

$$Q_\beta(y) = \log(\beta_{12}e^{y_1+y_2} + \beta_{13}e^{y_1} + \beta_{23}e^{y_2}),$$

and

$$\phi_\alpha(s) = \log h_\alpha(e^s).$$

Then

$$G_{\alpha,\beta}(y) := \log F_{\alpha,\beta}(x(y)) = \phi_\alpha(Q_\beta - 2L) + 3L - y_1 - y_2.$$

Lemma 5 (Elasticity bounds). *For $2/3 < \alpha < 1$ and $b > 0$,*

$$0 < E_\alpha(b) := \frac{bh'_\alpha(b)}{h_\alpha(b)} < \frac{3}{2}, \quad E'_\alpha(b) > 0.$$

Equivalently,

$$0 < \phi'_\alpha(s) < 3/2, \quad \phi''_\alpha(s) > 0.$$

Proof. With

$$u = \cos(\alpha\pi/2), \quad K = 1 - 4u^2,$$

parameterize the cubic boundary by

$$b = r(Kr - 2u), \quad h = r^2(1 + 2ur), \quad r > 2u/K.$$

Then

$$E = \frac{b}{h} \frac{dh/dr}{db/dr} = \frac{(Kr - 2u)(1 + 3ur)}{(Kr - u)(1 + 2ur)}.$$

The two factors

$$\frac{Kr - 2u}{Kr - u}, \quad \frac{1 + 3ur}{1 + 2ur}$$

increase strictly with r , while $E \rightarrow 0$ as $r \downarrow 2u/K$ and $E \rightarrow 3/2$ as $r \rightarrow \infty$. Since $b(r)$ is strictly increasing, the result follows. $\square$

Theorem 4 (Global strict logit convexity). *For every $\beta > 0$ and $2/3 < \alpha < 1$, $G_{\alpha,\beta}$ is globally strictly convex and coercive. Hence $F_{\alpha,\beta}$ has a unique nondegenerate minimizer*

$$x^*(\alpha, \beta) \in \Delta_2^\circ,$$

and $T_\alpha(\beta)$ and $x^(\alpha, \beta)$ depend smoothly on (α, β) .*

Proof. Writing $s = Q_\beta - 2L$,

$$\nabla^2 G = \phi''_\alpha(s) \nabla s \nabla s^\top + \phi'_\alpha(s) \nabla^2 Q_\beta + (3 - 2\phi'_\alpha(s)) \nabla^2 L.$$

Here $\nabla^2 Q_\beta \succeq 0$, while $\nabla^2 L \succ 0$ because L is the log-sum-exp of the three affinely independent vectors $(1, 0)$, $(0, 1)$, $(0, 0)$ with positive softmax weights. Lemma 5 makes every scalar coefficient nonnegative and the coefficient of $\nabla^2 L$ strictly positive. Thus

$$\nabla^2 G(y) \succ 0 \quad \forall y \in \mathbb{R}^2.$$

As $\|y\| \rightarrow \infty$, $x(y)$ approaches the simplex boundary, $x_1 x_2 x_3 \rightarrow 0$, and the numerator remains strictly positive; hence $G \rightarrow \infty$. Coercivity gives a unique global minimizer, positive definiteness gives nondegeneracy, and the implicit-function theorem yields smooth dependence. $\square$

5.2. Symmetric slice and canonical family

Corollary 4 (Fully symmetric slice). *If*

$$\beta_{12} = \beta_{13} = \beta_{23} = b > 0,$$

then

$$x^* = (1/3, 1/3, 1/3), \quad T_\alpha(b, b, b) = 27h_\alpha(b/3).$$

For $b = 1$,

$$T_\alpha(1, 1, 1) = 1 + R_3(\alpha)^3, \quad R_3(\alpha) = \frac{\sin(\alpha\pi/2)}{\sin(\alpha\pi/2 - \pi/3)}.$$

Proof. Permutation symmetry and uniqueness force x^* to be the simplex center. Substitution gives the first identity; the second simplifies to the $n = 3$ single-cycle threshold of Siami [8]. $\square$

Consider

$$A_\gamma = \begin{pmatrix} -1 & 0 & -\gamma \\ \gamma & -1 & 0 \\ 0 & \gamma & -1 \end{pmatrix}, \quad \gamma > 0.$$

It has

$$\beta = (1, 1, 1), \quad \kappa = 1 + \gamma^3.$$

Corollary 5 (Exact cyclic threshold). For $2/3 < \alpha < 1$,

$$A_\gamma \in \mathcal{F}_\alpha^{(3)} \iff \gamma < R_3(\alpha),$$

whereas

$$A_\gamma \in \mathcal{D}_H^{(3)} \iff \gamma < 2.$$

Thus

$$2 < \gamma < R_3(\alpha)$$

is an exact one-parameter genuinely fractional D -stable band.

Proof. The fractional statement follows from Theorem 2 and Corollary 4; the classical endpoint is

$$1 + \gamma^3 < T_1(1, 1, 1) = 9.$$

$\square$

5.3. Order dependence and the classical limit

Theorem 5 (Strict monotonicity). For fixed $\beta > 0$,

$$\frac{2}{3} < \alpha_1 < \alpha_2 \leq 1 \implies T_{\alpha_1}(\beta) > T_{\alpha_2}(\beta).$$

Proof. For fixed $a, b > 0$, the cubic boundary value $H_\theta(a, b)$ strictly decreases as the forbidden angle θ increases: a cubic touching the larger-angle ray lies strictly inside the wider Matignon region associated with the smaller angle. Hence

$$h_{\alpha_1}(b) > h_{\alpha_2}(b) \quad \forall b > 0.$$

Evaluating the two objectives at a minimizer of the α_1 problem gives the stated strict inequality. $\square$

Set

$$S = \sqrt{\beta_{12}} + \sqrt{\beta_{13}} + \sqrt{\beta_{23}}, \quad \Gamma = \sqrt{\beta_{12}\beta_{13}\beta_{23}}.$$

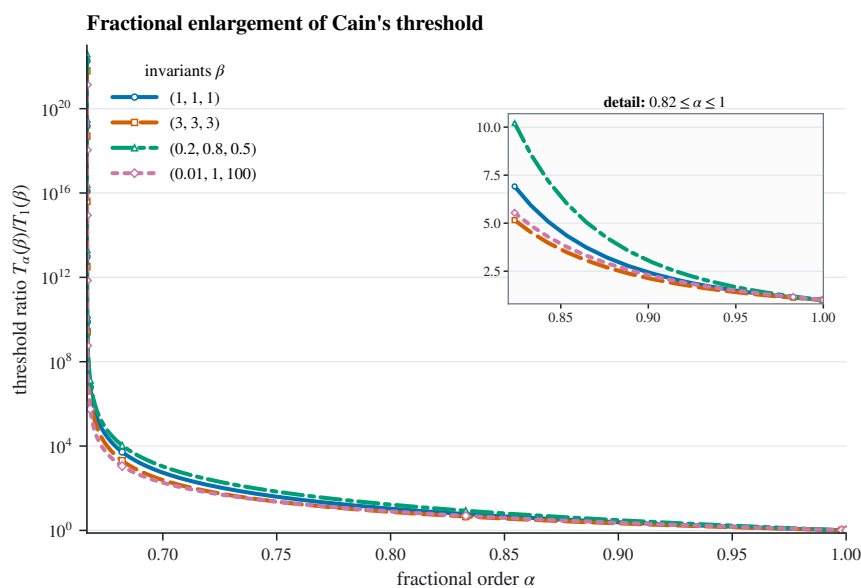


Figure 1. Fractional enlargement of Cain's threshold for representative invariant triples. Theorem 5 gives strict decrease in α , and Theorem 6 gives convergence to T_1 at integer order. The main ordinate is logarithmic; the inset resolves the approach to integer order on $0.82 \leq \alpha \leq 1$. Colors, line styles, and markers redundantly identify the four triples.

Theorem 6 (Sharp classical-limit rate). As $\alpha \rightarrow 1^-$,

$$T_\alpha(\beta) = T_1(\beta) + C(\beta)(1 - \alpha) + O((1 - \alpha)^2),$$

where

$$T_1(\beta) = S^2, \quad C(\beta) = \pi \left(\frac{S^{5/2}}{\sqrt{\Gamma}} + S^{3/2} \sqrt{\Gamma} \right).$$

Proof. Put $\varepsilon = 1 - \alpha$. Since

$$u = \cos(\alpha\pi/2) = \frac{\pi}{2}\varepsilon + O(\varepsilon^3),$$

implicit differentiation of the normalized boundary equation at $u = 0$ gives

$$h_\alpha(b) = b + \pi\varepsilon\sqrt{b}(1 + b) + O(\varepsilon^2).$$

At $\alpha = 1$ the unique simplex minimizer is

$$x_1^* = \frac{\sqrt{\beta_{23}}}{S}, \quad x_2^* = \frac{\sqrt{\beta_{13}}}{S}, \quad x_3^* = \frac{\sqrt{\beta_{12}}}{S},$$

with

$$x_1^* x_2^* x_3^* = \frac{\Gamma}{S^3}, \quad B_\beta(x^*) = \frac{\Gamma}{S}.$$

The envelope theorem, justified by Theorem 4, therefore gives

$$C(\beta) = \pi \frac{\sqrt{\Gamma/S}(1 + \Gamma/S)}{\Gamma/S^3},$$

which simplifies to the stated formula. $\square$

5.4. Minimum dimension for robust genuinely fractional separation

Theorem 7 (Minimum robust dimension). *For every $0 < \alpha < 1$,*

$$\min \left\{ n : \text{int} \left(\mathcal{F}_\alpha^{(n)} \setminus \mathcal{D}_H^{(n)} \right) \neq \emptyset \right\} = 3.$$

Proof. For $n = 1$, the two stability notions coincide. For $n = 2$, Corollary 1 gives empty interior.

For $n = 3$, use A_γ . It satisfies $-A_\gamma$ strict P for every $\gamma > 0$, and it is Hurwitz D-stable exactly for $\gamma < 2$. If $0 < \alpha \leq 2/3$, Kellogg's theorem gives $A_\gamma \in \mathcal{F}_\alpha^{(3)}$ for every γ , so any $\gamma > 2$ lies in the genuinely fractional class. If $2/3 < \alpha < 1$, Corollary 5 gives fractional D-stability for $\gamma < R_3(\alpha)$, and $R_3(\alpha) > 2$ because

$$T_\alpha(1, 1, 1) > T_1(1, 1, 1) = 9.$$

Choose $2 < \gamma < R_3(\alpha)$. In both regimes the defining inequalities are strict, so the difference class contains an open neighborhood of the chosen matrix. $\square$

The proof-complete technical version of the manuscript also develops a global parametrization of the threshold surface, exact invariant-space sensitivities, the $\alpha \downarrow 2/3$ asymptotic, a closed-form sufficient certificate, and a general realizability theorem. Those secondary results are retained in the accompanying technical manuscript and reproducibility archive but are not needed for the central elimination theorem or the minimum-dimension conclusion.

6. Three-species ecological interpretation

Consider the commensurate fractional GLV system

$${}^C D_t^\alpha x_i = x_i \left(r_i + \sum_j a_{ij} x_j \right),$$

a positive equilibrium $x^* \gg 0$ has Jacobian

$$J = \text{diag}(x^*)A.$$

The results below concern local Jacobian stability. Principal-minor and cycle decompositions are classical in ecological stability analysis [11,12]. Here the exact fractional all-diagonal boundary is written directly in those coordinates.

Proposition 1 (Positive-equilibrium orbit equivalence). *For every $x^* \gg 0$,*

$$J \in \mathcal{F}_\alpha^{(3)} \iff A \in \mathcal{F}_\alpha^{(3)}, \quad J \in \mathcal{D}_H^{(3)} \iff A \in \mathcal{D}_H^{(3)}.$$

Hence $J \in \mathcal{P}_\alpha^{(3)} \iff A \in \mathcal{P}_\alpha^{(3)}$, and the four invariants (β, κ) are unchanged.

Proof. If $E = \text{diag}(x^*)$, then

$$\{DJ : D \succ 0\} = \{DEA : D \succ 0\} = \{\tilde{D}A : \tilde{D} \succ 0\},$$

because $D \mapsto DE$ is a bijection of the positive diagonal cone. The invariant ratios are unchanged by the same row scaling. $\square$

Write

$$p_i = -a_{ii} > 0, \quad g_{ij} = \frac{a_{ij}a_{ji}}{p_i p_j}.$$

Then

$$\beta_{ij} = 1 - g_{ij}.$$

Thus opposite-sign reciprocal interactions have $g_{ij} < 0$ and increase β_{ij} , whereas same-sign reciprocal interactions have $g_{ij} > 0$ and decrease it, subject to strict P.

For the two oriented three-cycle products,

$$\ell_{123} = \frac{a_{12}a_{23}a_{31}}{p_1 p_2 p_3}, \quad \ell_{132} = \frac{a_{13}a_{32}a_{21}}{p_1 p_2 p_3},$$

put

$$L_3 = \ell_{123} + \ell_{132}.$$

Proposition 2 (Determinant-loop identity).

$$\kappa = \beta_{12} + \beta_{13} + \beta_{23} - 2 - L_3.$$

Proof. Expanding $\det A$ and dividing by $p_1 p_2 p_3$ gives

$$\frac{\det A}{p_1 p_2 p_3} = -1 + L_3 + g_{12} + g_{13} + g_{23}.$$

Use $\kappa = -\det A / (p_1 p_2 p_3)$ and $g_{ij} = 1 - \beta_{ij}$. $\square$

Corollary 6 (Exact loop-space fractional band). For $2/3 < \alpha < 1$, set

$$C_\beta = \beta_{12} + \beta_{13} + \beta_{23} - 2.$$

On the strict-P stratum,

$$A \in \text{int } \mathcal{P}_\alpha^{(3)}$$

if and only if

$$C_\beta - T_\alpha(\beta) < L_3 < C_\beta - T_1(\beta).$$

Proof. Substitute $\kappa = C_\beta - L_3$ into Corollary 3. $\square$

The ecological interpretation is therefore invariant under positive equilibrium abundance scaling. It separates normalized reciprocal-pair feedback $\beta_{ij} = 1 - g_{ij}$ from total directed three-cycle feedback L_3 . This is a structural local-stability statement; it does not assert equilibrium feasibility, global persistence, or a causal effect of changing one biological interaction coefficient in isolation.

7. Computational certification and reproducibility

All theorem claims above are analytic. Computation was used only as an independent falsification and certification layer. Two routes were kept separate: an invariant-space implementation of

$$T_\alpha(\beta) = \min_{x \in \Delta_2^\circ} \frac{h_\alpha(B_\beta(x))}{x_1 x_2 x_3},$$

and direct optimization of the spectral margin

$$\min_{D \succ 0} \left[\min_{\lambda \in \sigma(DA)} |\arg \lambda| - \frac{\alpha \pi}{2} \right].$$

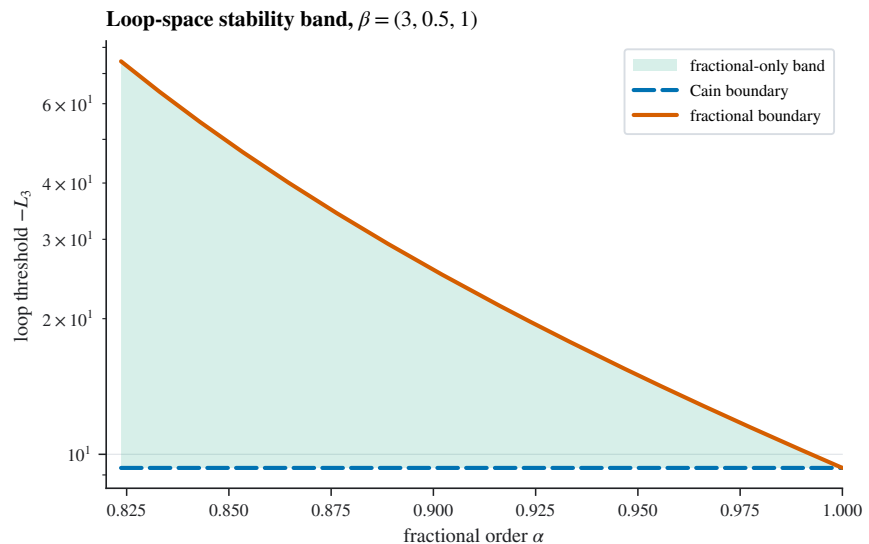


Figure 2. Cain and fractional loop-space boundaries for $\beta = (3, 0.5, 1)$. The shaded interval between the dashed Cain boundary and the solid fractional boundary is the genuinely fractional region; the ordinate is logarithmic.

The latter uses only eigenvalues of the full matrix DA , so agreement tests the elimination theorem rather than a second implementation of the same formula.

Across two campaigns, 2,180,000 adversarial matrix/order cases produced no genuine discrepancy with Theorem 2; all 56,234 numerically delicate cases were independently recomputed in multiprecision. A second atlas enclosed 938 threshold values by interval arithmetic. Of these, 903 had a global branch-and-bound certificate independent of Theorem 4; 35 extreme-anisotropy cases used the proved convexity/uniqueness result to complete localization. The worst relative interval width was 1.1×10^{-35} , and every independent 50-digit optimizer lay inside its reported enclosure.

The 35-matrix canonical library includes symmetric and nonsymmetric examples, matrices near the Cain and fractional boundaries, mixed-sign ecological examples, and matrices that are Hurwitz at $D = I$ but not Hurwitz D -stable. One representative is

$$A = \begin{pmatrix} -1 & -3/2 & 0 \\ -1/2 & -1 & -3 \\ -3 & 0 & -2 \end{pmatrix}, \quad \alpha = 0.7.$$

Here

$$\beta = (1/4, 1, 1), \quad \kappa = 7, \quad T_1 = 6.25, \quad T_\alpha \approx 4726.94.$$

Although

$$\max \Re \sigma(A) \approx -0.0789,$$

the Cain-minimizing positive diagonal gives

$$\max \Re \sigma(D_H A) \approx 0.0384 > 0.$$

Thus A itself is Hurwitz, its classical positive-diagonal orbit is not, but its full orbit is Matignon-stable at $\alpha = 0.7$.

The differential structure of the threshold was also checked at 8,000 parameter points. The 500 worst floating-point cases were recomputed at 40 digits; the identity

$$\frac{\partial T_\alpha}{\partial \beta_{ij}} = \frac{h'_\alpha(B^*)}{x_k^*}$$

then agreed to at worst 2.4×10^{-33} , and the logit Hessian remained positive definite in every tested case. The final repository test suite has 135 passing tests and no failures.

All scripts, fixed seeds, software metadata, interval certificates, canonical matrices, and figure-source data are maintained in the public repository

<https://github.com/maximilianolucius/fractional-d-stability-networks>.

The numerical evidence is reproducible but logically secondary: no theorem is inferred from finite sampling.

8. Discussion

The main result is an explicit solution of a known generalized-D-stability problem in the first dimension where a robust genuinely fractional separation is possible. Generalized multiplicative region stability is not new [2]; Kushel and Pavani [3] give an abstract all-positive-diagonal forbidden-boundary framework. Our contribution is the exact elimination of that diagonal quantifier for the generic real 3×3 Matignon problem on the full-dimensional strict- $P(-A)$ stratum.

The closest classical predecessor is Cain's complete real- 3×3 Hurwitz D-stability criterion [1]. Here it appears exactly as the endpoint

$$T_1(\beta) = \left(\sqrt{\beta_{12}} + \sqrt{\beta_{13}} + \sqrt{\beta_{23}} \right)^2,$$

while Theorem 5 shows that $T_\alpha > T_1$ for every fractional order in the high-order regime. Bahl–Cain fixed-inertia theory [13] is complementary but cannot determine Matignon stability because the fractional stable region includes a right-half-plane sliver and therefore requires angular information finer than inertia. The cyclic secant theorem of Siami [8] is recovered exactly as the symmetric invariant slice.

Classical sufficient computational tests, including LMI approaches [14], and scalable determinant and principal-minor conditions [15,16] address a different tradeoff: tractable sufficient testing in general dimension rather than an exact angular elimination in dimension three.

Theorem 7 explains why $n = 3$ is not merely a convenient small example. In dimension two the fractional-only class is lower-dimensional; in dimension three it has nonempty interior for every $0 < \alpha < 1$. The positive-diagonal orbit reduction extends to higher dimension, but from $n = 4$ onward the normalized polynomial contains several independent angular coefficients, so the cubic scalar-boundary argument no longer transfers directly. Structured 4×4 families are therefore reserved for future work.

The strict-P hypothesis should also be interpreted correctly. Corollary 2 shows that every full-dimensional interior point lies on that stratum, so the theorem completely describes the robust class. It does not classify every lower-dimensional boundary component with vanishing signed principal minors. This distinction parallels classical work on the interior and strong D-stability [17,18].

Finally, the GLV interpretation is local. Positive equilibrium abundances reparameterize the same positive-diagonal orbit, and the loop variables give invariant coordinates for that orbit. They do not by themselves imply feasibility, global persistence, nonlinear basin stability, or a one-parameter causal intervention rule.

The principal limitations are therefore explicit: common fractional order, real matrices, full-dimensional classification, dimension three, and local equilibrium stability. These boundaries define the natural continuation of the theory rather than qualifications of the exact three-dimensional result.

9. Conclusion

For real 3×3 matrices on the full-dimensional strict- $P(-A)$ stratum, positive-diagonal Matignon stability is exactly equivalent to

$$\kappa < T_\alpha(\beta),$$

where four invariant principal-minor ratios reduce the entire positive-diagonal orbit to a two-dimensional simplex. The threshold recovers Cain's classical D-stability surface at $\alpha = 1$, is globally strictly convex after a logit transformation, and has a unique smooth optimizer.

The strict band

$$T_1(\beta) < \kappa < T_\alpha(\beta)$$

therefore gives an exact orbit-level separation between fractional and Hurwitz D-stability. Together with the complete 2×2 classification and the explicit cyclic family, this shows that dimension three is the smallest real dimension in which the difference has nonempty full-dimensional interior for every $0 < \alpha < 1$.

For three-species generalized Lotka–Volterra systems, positive equilibrium abundance scaling does not change the positive-diagonal orbit. The exact threshold can therefore be expressed in abundance-independent reciprocal-pair and directed-cycle feedback coordinates. Independent high-precision and interval computations found no discrepancy with the analytic theory and provide reproducible certified examples.

The general simplex reduction persists beyond dimension three, but the higher-order angular geometry does not collapse to the same scalar cubic boundary. Extending exact positive-diagonal Matignon stability to structured 4×4 families is the next natural problem.

Author Contributions: Conceptualization, methodology, formal analysis, investigation, writing—original draft, writing—review and editing, I.A. and T.N.A.; software and validation, I.A. and T.N.A. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: All computational scripts and reproducibility materials used in this study are included in the accompanying submission materials and project repository.

Conflicts of Interest: The authors declare no conflicts of interest.

Declaration on Generative AI Use

Generative AI tools, including OpenAI ChatGPT, were used to assist with drafting and editing, organization of literature searches, mathematical proof exploration and consistency checking, development and review of symbolic and numerical code, and preparation of reproducibility materials. These tools are not authors. The authors assume responsibility for all content.

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